

CALCULUS WITH

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MIDSEM RECAP | TOPIC SERIES

Midsem Recap

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MATH 152 | Calculus & Analysis | KNUST BME1

Read This First

These eight questions are reconstructed from memory — close paraphrases of what appeared on **this year's midsem**, not an official release or answer sheet.

Some of these ideas have a decent chance of resurfacing on the final, chale — but that's no guarantee any specific question repeats. What actually carries over is the **reasoning**: learn the concept, not the answer key.

Each question below is followed by a **step-by-step solution** and a **What This Tests** box explaining the concept. Read both, even if you got it right.

Final Exam Format

Reconstructed from the lecturer's stated pattern and this year's midsem. **Subject to change** — treat this as a planning guide, not a guarantee.



PART 1 --- Q1--15, MCQ

Integration by parts, substitution, and partial fractions; hyperbolic, logarithmic, inverse-hyperbolic, trigonometric, and inverse-trigonometric integrals; plus derivatives, L'Hôpital's Rule, implicit differentiation, and trigonometric/hyperbolic derivatives.

PART 2 --- Q16--25, FILL-IN-THE-BLANK

Rolle's Theorem, the Mean Value Theorem, the Intermediate Value Theorem, the Bolzano–Weierstrass Theorem, the Extreme Value Theorem, and the formal definitions of Cauchy, convergent, and divergent sequences.

PART 3 --- Q26--35, MCQ

Limit points, open and closed intervals, bounded sets, Cauchy sequences, limits involving e and $\ln$, and piecewise limits.

This structure lines up with what Part 1's questions 1–4 and 7 in this booklet look like, Part 2 with question 8, and Part 3 with questions 2, 5, and 6 — so working through this recap touches all three parts of the real exam.

1. Limit at Infinity --- Exponential vs. Polynomial

EXAM QUESTION

Evaluate the following limit:

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^5}$$

- (A) 0
- (B) 1
- (C) ∞
- (D) $\frac{1}{120}$

Correct answer: (C)

STEP-BY-STEP SOLUTION

This is an $\frac{\infty}{\infty}$ indeterminate form, so apply L'Hôpital's Rule — repeatedly, since the denominator is a degree-5 polynomial:

$$\lim_{x \rightarrow \infty} \frac{e^x}{x^5} \rightarrow \frac{e^x}{5x^4} \rightarrow \frac{e^x}{20x^3} \rightarrow \frac{e^x}{60x^2} \rightarrow \frac{e^x}{120x} \rightarrow \frac{e^x}{120}$$

Each application differentiates numerator and denominator once. After five applications the denominator becomes the constant 120, while the numerator is still e^x , which grows without bound. So the limit is ∞ .

WHAT THIS TESTS

The core idea: exponential growth always outpaces polynomial growth, no matter how high the polynomial's degree. You're also being tested on whether you know L'Hôpital's Rule can be applied more than once in a row when each application still leaves an indeterminate form — and on not giving up after the first pass.

2. Sequence Convergence

EXAM QUESTION

Consider the alternating real sequence: $2, -2, 2, -2, 2, -2, \dots$
Which of the following properties correctly describes this sequence?

- (A) It is a convergent sequence.
- (B) It is a Cauchy sequence.
- (C) It is a divergent sequence.
- (D) It converges to 0.

Correct answer: (C)

STEP-BY-STEP SOLUTION

A sequence converges only if its terms eventually settle arbitrarily close to one single value. Here the terms never settle — they permanently alternate between 2 and -2 , so no matter how far out you go, consecutive terms stay exactly 4 apart. There is no candidate limit L that every term gets close to, so the sequence is divergent. It also cannot be Cauchy, since a Cauchy sequence requires terms to get arbitrarily close to *each other* eventually — these never do (they're always 4 apart).

WHAT THIS TESTS

Whether you can tell the difference between a sequence that's bounded and one that's convergent — this sequence is bounded (it never leaves $[-2, 2]$) but still diverges, because boundedness alone doesn't guarantee a limit. It's also testing the Cauchy definition directly: Cauchy and convergent are equivalent for real sequences, so ruling out convergence rules out Cauchy too.

3. Range of a Piecewise Function

EXAM QUESTION

Determine the complete range of the piecewise defined function:

$$f(x) = \begin{cases} x^2 & x < 1 \\ 2x - 1 & x \geq 1 \end{cases}$$

- (A) $(-\infty, \infty)$
- (B) $[0, \infty)$
- (C) $[1, \infty)$
- (D) $(0, 1)$

Correct answer: (B)

STEP-BY-STEP SOLUTION

Handle each piece's output set separately, then union them.

Piece 1 ($x < 1$): x ranges over all reals less than 1, including arbitrarily negative values. Squaring a very negative number still gives a very large positive number, so x^2 is *not* bounded above here. The minimum of x^2 on this piece occurs at $x = 0$ (which is allowed, since $0 < 1$), giving 0. So this piece's output is $[0, \infty)$.

Piece 2 ($x \geq 1$): $2x - 1$ is linear and increasing, starting at $2(1) - 1 = 1$ and increasing without bound. So this piece's output is $[1, \infty)$.

Union: $[0, \infty) \cup [1, \infty) = [0, \infty)$.

WHAT THIS TESTS

The classic trap is assuming " $x < 1$ " restricts the *output* the way it restricts the input — students often answer $(0, 1)$ or $[1, \infty)$ by only checking values of x near the boundary, forgetting the piece extends to $-\infty$. Always check what happens at the extreme ends of each piece's domain, not just near the seam.

4. Continuity vs. Differentiability

EXAM QUESTION

Analyze the function $f(x)$ at the point $x = 0$:

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

- (A) Differentiable but not continuous at $x = 0$.
- (B) Neither continuous nor differentiable at $x = 0$.
- (C) Continuous and differentiable at $x = 0$.
- (D) Continuous but not differentiable at $x = 0$.

Correct answer: (D)

STEP-BY-STEP SOLUTION

Continuity: $\sin(1/x)$ oscillates between -1 and 1 but never leaves that range, so $-|x| \leq x \sin(1/x) \leq |x|$ for all $x \neq 0$. Both bounds $\rightarrow 0$ as $x \rightarrow 0$, so by the Squeeze Theorem $\lim_{x \rightarrow 0} x \sin(1/x) = 0 = f(0)$. f is continuous at 0 .

Differentiability: check the difference quotient

$$\frac{f(h) - f(0)}{h} = \frac{h \sin(1/h)}{h} = \sin\left(\frac{1}{h}\right)$$

As $h \rightarrow 0$, $1/h \rightarrow \pm\infty$, and $\sin(1/h)$ oscillates forever between -1 and 1 without settling on any value. The limit does not exist, so f is *not* differentiable at $x = 0$.

WHAT THIS TESTS

This is the Squeeze Theorem and the definition of the derivative, deployed together. It's also the same underlying idea as $|x|$ being continuous but not differentiable at 0 — except here the failure isn't a sharp corner, it's a slope that never settles down at all because of the oscillation. Whenever you see a $\sin(1/x)$ -type term multiplied by something that shrinks to 0 , expect a Squeeze Theorem argument.

5. Trigonometric Limit --- Taylor-Series Flavour

EXAM QUESTION

Evaluate the trigonometric limit:

$$\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$$

- (A) 0
- (B) $\frac{1}{3}$
- (C) 1
- (D) The limit does not exist (DNE).

Correct answer: (B)

STEP-BY-STEP SOLUTION

Direct substitution gives $\frac{0}{0}$, and it stays $\frac{0}{0}$ even after one application of L'Hôpital — so apply it repeatedly:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan x - x}{x^3} &\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{\sec^2 x - 1}{3x^2} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{2 \sec^2 x \tan x}{6x} \\ &\stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{2 \sec^4 x + 4 \sec^2 x \tan^2 x}{6} \end{aligned}$$

Substituting $x = 0$ into the last expression: $\sec(0) = 1$, $\tan(0) = 0$, giving

$$\frac{2(1) + 4(1)(0)}{6} = \frac{2}{6} = \frac{1}{3}.$$

WHAT THIS TESTS

Stamina with repeated L'Hôpital applications, plus careful differentiation of $\tan x$ and $\sec x$ at each stage. This result — $\frac{1}{3}$ — also matches what you'd get from the Taylor expansion $\tan x \approx x + \frac{x^3}{3} + \dots$, so if that's covered in your course, it's a useful cross-check.

6. Sequence Limit --- The Definition of e

EXAM QUESTION

Evaluate the following logarithmic sequence limit:

$$\lim_{n \rightarrow \infty} \ln \left[\left(1 + \frac{1}{n} \right)^n \right]$$

- (A) 0
- (B) 1
- (C) e
- (D) ∞

Correct answer: (B)

STEP-BY-STEP SOLUTION

Recall the defining limit of Euler's number:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

Since $\ln$ is continuous everywhere it's defined (in particular, continuous at e), you can pass the limit inside the logarithm:

$$\lim_{n \rightarrow \infty} \ln \left[\left(1 + \frac{1}{n} \right)^n \right] = \ln \left(\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n \right) = \ln(e) = 1$$

WHAT THIS TESTS

Whether you recognise $\left(1 + \frac{1}{n} \right)^n$ on sight as the definition of e , rather than trying to brute-force it with L'Hôpital on an ∞^0 -type indeterminate form. It's also testing that you know continuous functions commute with limits: $\lim f(a_n) = f(\lim a_n)$ when f is continuous at the limit point.

7. Extreme Values on a Closed Interval

EXAM QUESTION

Find the absolute maximum and absolute minimum values of the function $f(x) = x^3 - 3x$ on the closed interval $[-2, 2]$.

- (A) Max = 2, Min = -2
- (B) Max = 4, Min = -4
- (C) Max = 1, Min = -1
- (D) Max = 18, Min = -18

Correct answer: (A)

STEP-BY-STEP SOLUTION

By the Extreme Value Theorem, a continuous function on a closed interval attains its max and min — either at a critical point or at an endpoint. Find critical points first:

$$f'(x) = 3x^2 - 3 = 0 \implies x^2 = 1 \implies x = \pm 1$$

Both $x = 1$ and $x = -1$ lie inside $[-2, 2]$, so evaluate f at all critical points *and* both endpoints:

$$f(-2) = -8 + 6 = -2 \qquad f(-1) = -1 + 3 = 2$$

$$f(1) = 1 - 3 = -2 \qquad f(2) = 8 - 6 = 2$$

The largest value is 2 (attained at both $x = -1$ and $x = 2$); the smallest is -2 (attained at both $x = -2$ and $x = 1$).

WHAT THIS TESTS

The full Extreme-Value-Theorem procedure: find every critical point in the interval, then compare the function's value there against *both* endpoints — not just one. A common mistake is checking only the critical points, or only one endpoint, and missing that the true max/min can tie between an endpoint and an interior point.

8. Bolzano--Weierstrass Theorem

EXAM QUESTION

Which foundational calculus theorem states that a bounded sequence of real numbers must contain a convergent subsequence?

- (A) Rolle's Theorem
- (B) Mean Value Theorem (MVT)
- (C) Intermediate Value Theorem (IVT)
- (D) Bolzano--Weierstrass Theorem (BWT)

Correct answer: (D)

STEP-BY-STEP SOLUTION

This one is definition-recall rather than computation, so the “solution” is knowing what each theorem actually claims:

- **Rolle's Theorem:** if $f(a) = f(b)$ for a differentiable f , some $c \in (a, b)$ has $f'(c) = 0$.
- **MVT:** some $c \in (a, b)$ has $f'(c)$ equal to the average slope over $[a, b]$.
- **IVT:** a continuous function on $[a, b]$ hits every value between $f(a)$ and $f(b)$.
- **Bolzano--Weierstrass:** *every bounded sequence of real numbers has a convergent subsequence* — this matches the question exactly.

None of the other three theorems say anything about sequences at all — they're all about derivatives or continuity of functions on an interval, which immediately rules out (A), (B), and (C).

WHAT THIS TESTS

Straight recall, but recall that requires you to actually distinguish four theorems that get blurred together because they're taught in the same stretch of the course. The reliable filter: Rolle's/MVT/IVT are all about a *function on an interval*; Bolzano--Weierstrass is the only one of the four about a *sequence*. If the question mentions “sequence,” BWT is the only real candidate.